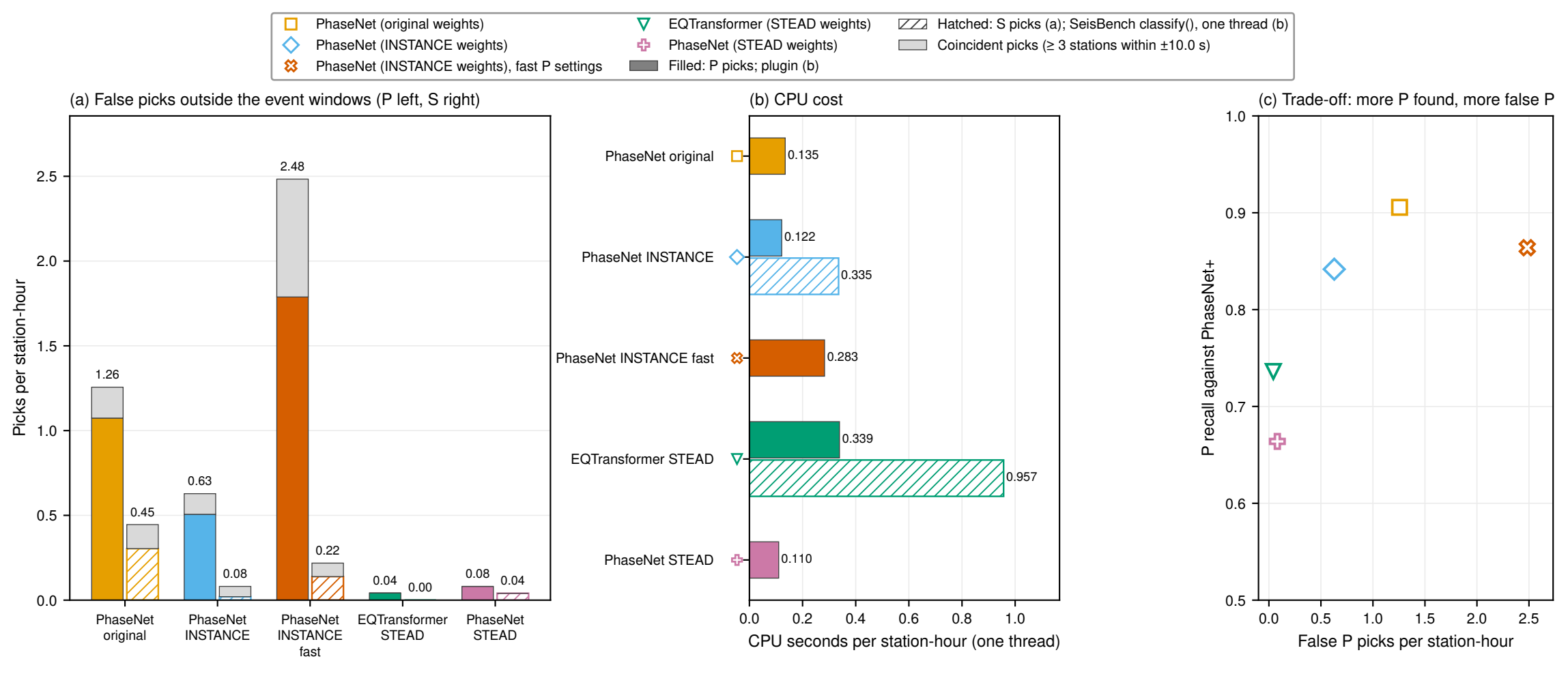
PhaseNet (original weights)
PhaseNet (INSTANCE weights)
PhaseNet (INSTANCE weights), fast P settings
EQTransformer (STEAD weights)
PhaseNet (STEAD weights)
Filled: P picks; plugin (b)
Hatched: S picks (a); SeisBench classify(), one thread (b)
Coincident picks (≥ 3 stations within ±10.0 s)
(a) False picks outside the event windows (P left, S right)
Picks per station-hour
1.26
0.45
0.63
0.08
2.48
0.22
0.04
0.00
0.08
0.04
PhaseNet original
PhaseNet INSTANCE
PhaseNet INSTANCE fast
EQTransformer STEAD
PhaseNet STEAD
(b) CPU cost
PhaseNet original
PhaseNet INSTANCE
PhaseNet INSTANCE fast
EQTransformer STEAD
PhaseNet STEAD
0.135
0.122
0.335
0.283
0.339
0.957
0.110
CPU seconds per station-hour (one thread)
(c) Trade-off: more P found, more false P
P recall against PhaseNet+
False P picks per station-hour